

MATH 3210-004 PSet 8

Specification

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1 Background

This problem set focuses on differential calculus. The first three problems are on **orders of magnitude** as well as various foundational aspects of derivatives, while the remaining problems explore the consequence of the mean value theorem.

Tangency Let X be a metric space, $x_* \in X$, and $\beta \in F(\mathbb{R}; \mathbb{R}_{\geq 0})$. Then two functions $f, g \in F(X; \mathbb{R})$ are said to be **order- β tangent at x_*** if any one of the following equivalent statements hold:

- $f(x) = g(x) + o_{x \rightarrow x_*}(\beta \circ d(x, x_*))$.
- $\lim_{x \rightarrow x_*} \frac{|f(x) - g(x)|}{\beta \circ d(x, x_*)} = 0$.
- $\forall \varepsilon \in \mathbb{R}_{>0}, \exists \delta \in \mathbb{R}_{>0}, \forall x \in X : d(x, x_*) < \delta \implies |f(x) - g(x)| < \varepsilon \beta \circ d(x, x_*)$.

If X is an open interval, then the above conditions can be rewritten as follows (using the change of variables $x = x_* + h$):

- $f(x_* + h) = g(x_* + h) + o_{h \rightarrow 0}(\beta(h))$.
- $\lim_{h \rightarrow 0} \frac{|f(x_* + h) - g(x_* + h)|}{\beta(h)} = 0$.
- $\forall \varepsilon \in \mathbb{R}_{>0}, \exists \delta \in \mathbb{R}_{>0}, \forall x \in X : |h| < \delta \implies |f(x_* + h) - g(x_* + h)| < \varepsilon \beta(h)$.

We write $f \parallel_{x_*, \beta} g$ to denote f and g being order- β tangent at x_* . We say f and g are **tangent at** x_* if f and g being order- β tangent at x_* for $\beta : x \mapsto x$, and in this case we write $f \parallel_{x_*} g$.

Equivalence Relations Let X be a set. A relation R on X is called an **equivalence relation** if

Reflexivity: $\Delta_X \subseteq R$, where $\Delta_X = \{(x_1, x_2) \in X \times X \mid x_1 = x_2\}$ is the diagonal relation¹,

Symmetry: $R^{-1} = R$ ², and

Transitivity: $R \circ R = R$ ³.

Antiderivatives and Newton Integrability Let I be a nonempty open interval, $f, g \in F(I; \mathbb{R})$. g is said to be an **antiderivative** of f if g is differentiable and for any $x \in I$, $g'(x) = f(x)$.

f is said to be **Newton integrable** if it has an antiderivative⁴. We denote by $N(I; \mathbb{R})$ the collection of all Newton integrable functions, and

¹More explicitly, this means that for any $x \in X$, $(x, x) \in R$.

²More explicitly, this means that for any $x_1, x_2 \in X$, if $(x_1, x_2) \in R$, then $(x_2, x_1) \in R$.

³More explicitly, this means that for any $x_1, x_2, x_3 \in X$, if $(x_1, x_2) \in R$ and $(x_2, x_3) \in R$, then $(x_1, x_3) \in R$.

⁴This notion of integrability seemingly is the most pragmatic integration theory, say from the Calc I perspective: one *dares* integrating only functions that are derivatives, and through the Fundamental Theorem of Calculus. After all, whatever integration is, it ought to invert differentiation. While for functions often encountered in Calc I, it's fairly straightforward, albeit at times rather tricky, to compute an antiderivative, in general guessing an antiderivative is **not an easy task**!

for any $f \in N(I; \mathbb{R})$ and for any compact subinterval $K \subseteq I$, we define the **(unsigned, definite) Newton integral** of f on K by

$$\int_K f(x) dx = F(\max(K)) - F(\min(K)),$$

where F is any antiderivative of f . Note that as any two antiderivatives of the same function defined on an open interval differ by a constant⁵, the Newton integral of a Newton integrable function is independent of the choice of the antiderivative F used on the RHS.

2 Generative AI and Computer Algebra Systems Regulations

This section applies only if you choose to use either a **generative AI tool** (e.g. chatbots) or a **computer algebra system (CAS)** while working on this problem set. If you do not use such tools, you may skip this section.

The use of these tools while completing this problem set is permitted, provided it is done responsibly and in a manner that supports your learning. Note however that in exams you will not be allowed to use any AI or CAS.

2.1 Disclosure

If you use such tools, you must disclose this fact in the designated section of the form you will complete as part of your submission. You must also share your reflections on which parts of the problem required your own mathematical judgment, insight, or decision-making,

⁵Indeed, if F_1 and F_2 are both antiderivatives of the same function, then $(F_1 - F_2)' = 0$, and hence by a **result proved in class**, $F_1 - F_2$ is constant.

and which parts could reasonably be delegated to a computational or generative tool.

2.2 Guidelines for Responsible Use

If you use a generative AI tool or chatbot, you must adhere to the following guidelines:

- During the chat you may share with the chatbot parts or all of this specification document, as well as parts of the textbook or other sources.
- Directly asking the tool for complete problem solutions is prohibited.
- Appropriate uses include asking for conceptual explanations, checking intermediate steps, clarifying definitions or theorems, or exploring alternative solution approaches, provided that the final work submitted is your own.
- Regardless of the tools used, you are expected to understand, justify, and be able to reproduce all submitted work independently. The use of AI or CAS does not reduce or replace this expectation.

3 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details.

When tackling statements in MATH 3210, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example, you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook. Such correspondences will not be mentioned; it'll be up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Let I be an open interval, $x_* \in I$ and $f \in F(I; \mathbb{R})$. Then
 - (a) If $f(x_* + h) = O_{h \rightarrow 0}(|h|^2)$, then $f(x_* + h) = o_{h \rightarrow 0}(|h|)$.
 - (b) If $f(x_* + h) = o_{h \rightarrow 0}(|h|)$, then $f(x_* + h) = O_{h \rightarrow 0}(|h|^2)$.
 - (c) If $\alpha, \beta \in \mathbb{R}_{\geq 0}$ are with $\alpha \geq \beta$, and $f(x_* + h) = O_{h \rightarrow 0}(|h|^\alpha)$, then $\frac{f(x_* + h)}{(x_* + h)^\beta} = O_{h \rightarrow 0}(|h|^{\alpha-\beta})$.
2. Let I be an open interval, and $x_* \in I$. Then
 - (a) For any $\beta \in F(\mathbb{R}; \mathbb{R}_{\geq 0})$, the relation $\|_{x_*, \beta}$ of "being order- β tangent at x_* " is an equivalence relation on $F(I; \mathbb{R})$.
 - (b) If $f, g \in F(I; \mathbb{R})$ are tangent at x_* , then $f(x_*) = g(x_*)$.
 - (c) For any $f \in F(I; \mathbb{R})$, f is differentiable at x_* iff there is exactly one polynomial function of degree at most 1 tangent to f at x_* .
 - (d) For any $f \in F(I; \mathbb{R})$ and for any $n \in \mathbb{Z}_{\geq 0}$, there is at most one polynomial function of degree at most n that is order- β_n tangent to f at x_* , where $\beta_n : h \mapsto h^n$.

- (e) For any $f \in F(I; \mathbb{R})$ and for any $n \in \mathbb{Z}_{\geq 0}$, if $f : I \rightarrow \mathbb{R}$ is n -times differentiable at x_* , then for any n -times differentiable $g \in F(I; \mathbb{R})$ that is order- β_n tangent to f at x_* , where $\beta_n : h \mapsto h^n$, for any $k \in \{0, 1, \dots, n\}$: $f^{(k)}(x_*) = g^{(k)}(x_*)$.
3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function and $x_* \in \mathbb{R}$. For any $M \in \mathbb{R}_{>0}$, define the **zoom** of f centered at x_* with magnification factor M to be the function

$$\mathcal{Z}_{x_*, M}(f) : \mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto M \left(f \left(x_* + \frac{x}{M} \right) - f(x_*) \right).$$

Then:

- (a) If f is differentiable at x_* , then as $M \rightarrow \infty$, $\mathcal{Z}_{x_*, M}(f)$ converges pointwise to the function whose graph is the line through the origin with slope $f'(x_*)$.
- (b) If f is differentiable at x_* , then as $M \rightarrow \infty$, $\mathcal{Z}_{x_*, M}(f)$ converges uniformly to the function whose graph is the line through the origin with slope $f'(x_*)$.
- (c) If f is such that $\mathcal{Z}_{x_*, M}(f)$ converges pointwise as $M \rightarrow \infty$, then f is differentiable at x_* .
4. Let I be a nonempty open interval, $f \in F(I; \mathbb{R})$ and $n \in \mathbb{Z}_{\geq 0}$. Then f is a polynomial function of degree at most n iff f is $(n+1)$ times differentiable and for any $x \in I$, $f^{(n+1)}(x) = 0$.
5. Let I be a nonempty open interval, and $f \in F(I; \mathbb{R})$. Then
- (a) If f is Newton integrable, then it is continuous.
- (b) If f is Newton integrable, then it takes any value between any two values it takes (More explicitly, for any subinterval $J \subseteq I$, the image $f(J)$ is an interval).

- (c) If f is Newton integrable and $f^{-1}(0) = \emptyset$, then either for any $I = f^{-1}(\mathbb{R}_{>0})$ xor $I = f^{-1}(\mathbb{R}_{<0})$.
- (d) If f takes any value between any two values it takes, then it is Newton integrable.

6. Let I be a nonempty open interval and $f \in F(I; \mathbb{R})$. Then

- (a) f is locally Bourbaki integrable iff for any $x_* \in I$, the following one-sided limits exist as real numbers:

$$\lim_{\substack{x \rightarrow x_* \\ x < x_*}} f(x), \quad \lim_{\substack{x \rightarrow x_* \\ x > x_*}} f(x)$$

- (b) If f is Newton integrable, then f is locally Bourbaki integrable iff f is continuous.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/cr4ahQFXtFgb6b1S8>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

<https://www.gradescope.com/courses/1212058/assignments/7390766>,

see the Gradescope [documentation](#) for instructions.

5 When to Submit

This problem set is due on March 27, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.